



Offline Relink-Candidate Analysis — discriminability & pre-loss kinematics

Offline study of relink discriminability on **MOT17 train, SDP**, `mamba_whole_graph`, built from a **relink-OFF / interpolation-OFF** substrate so every track death and birth is

raw and every (lost → candidate) pair can be enumerated and GT-labelled.

This **reconciles** (does not simply overturn) the headline of `bidir_relink_data_analysis.md` ("pure geometry AUC ≈ 0.55 ").

The two numbers measure **different pools** and are both correct:

- On the **full** offline candidate pool, `bridge_dist` is a strong ranker (**AUC ≈ 0.895**) — but that power is almost entirely *rejecting candidates that are obviously too far away*.
- Restricted to the **gate's operating region** (`bridge_dist  $\leq 1$` , i.e. the spatially plausible candidates the online relink actually weighs), AUC collapses to **~ 0.65** ($\leq 0.5 \rightarrow 0.64$). Among genuine near look-alikes, geometry barely separates — exactly the old doc's point.

So: the full-pool AUC is useful for **negative-pool reduction (recall side)**, but the current

bridge is **not strong on the hard cases**, and the precision ceiling is **base rate**, not a

better residual. (An earlier draft of this doc over-claimed the 0.55 was pure "selection bias";

the honest framing is easy-pool vs hard-pool AUC.)

- **Date:** 2026-06-09
- **Substrate run:** `mamba_whole_graph --detector SDP --no-interpolate-tracklets`
- **Scripts:** `scripts/tools/build_relink_candidates.py`,

`analyze_preloss_motion.py` ,

`analyze_turn_baseline.py` , `sweep_speed_turn.py`

- **Artifacts:** `scripts/tools/out/relink_candidates.csv` , `speed_turn_sweep.npz` , figures below.

1. The no-relink / no-interp substrate

All three relink paths are off by default in this preset and were confirmed inactive:

`reid_mode=off` $\Rightarrow$ `use_semantic_mode=False` $\Rightarrow$ no semantic/motion relinker is built;

`relink_enabled` and `relink_bridge_enabled` default `False` . Interpolation is additionally

disabled so short within-id gaps are *not* filled — exposing every real track break.

Run	IDF1	MOTA	HOTA	AssA	IDs	FP	FN
relink-off, interp-on	73.0	76.8	66.5	63.8	558	4092	21387
relink-off, interp-off (substrate)	72.8	75.9	65.9	63.4	695	1166	25186

IDs **558** $\rightarrow$ **695**: turning interpolation off exposes ~ 137 extra death/birth events — exactly

the relink opportunities to study. Per-frame dump in `results/MOT17_eval/*.txt` .

2. Candidate dataset & rules

`build_relink_candidates.py` $\rightarrow$ `relink_candidates.csv` (**21,261 pairs, 27 cols**).

Per lost track A $\rightarrow$ candidate B:

1. **causality** — `B.first_frame > A.last_frame` (only link IDs born after A is lost)
2. **no temporal overlap** — `frames(A)  $\cap$  frames(B) =  $\emptyset$` (co-existing $\Rightarrow$ different people)
3. **gap gate** — `1  $\leq$  gap  $\leq$  300` (wide)

4. **spatial gate** — foot-dist / mean-height (default off / wide)

Uniqueness ("an ID cannot be linked twice") is *not* a hard filter — it is two tag columns `accepted` / `already_linked` from a greedy pass (smallest `bridge_dist` wins, which

mirrors `relink_bidir_propose_kernel` 's propose/commit semantics). Each pair carries GT labels

(`gt_lost/gt_cand/gt_match/gt_valid`) and geometric/motion features

(`gap, dist_h, fwd_resid, bwd_resid, bridge_dist, dir_cos, speed_h`).

The main tracker association is an **auction** solver

(`parallel_auction_shmem_kernel` /

`AuctionAlgorithm::Solve`). The **bridge relink** is *not* auction — it is greedy highest-score propose/commit, so the offline uniqueness tag matches it intentionally.

3. Bridge discriminability — separates well, base-rate kills precision

On the 19,615 GT-valid pairs (**256 true relink chances** vs 19,359 false):

- **AUC(`bridge_dist`) = 0.895**. Positive median 0.79 vs negative median 4.90 — nearly disjoint. AUC by gap-bin: `1-10` →0.962, `11-30` →0.891, `31-60` →0.867, `61-150` →0.859, `151-300` →0.836 (velocity-extrapolation reliability decays with gap).
- **But base rate is 1.3%**, so precision is low at any useful recall:

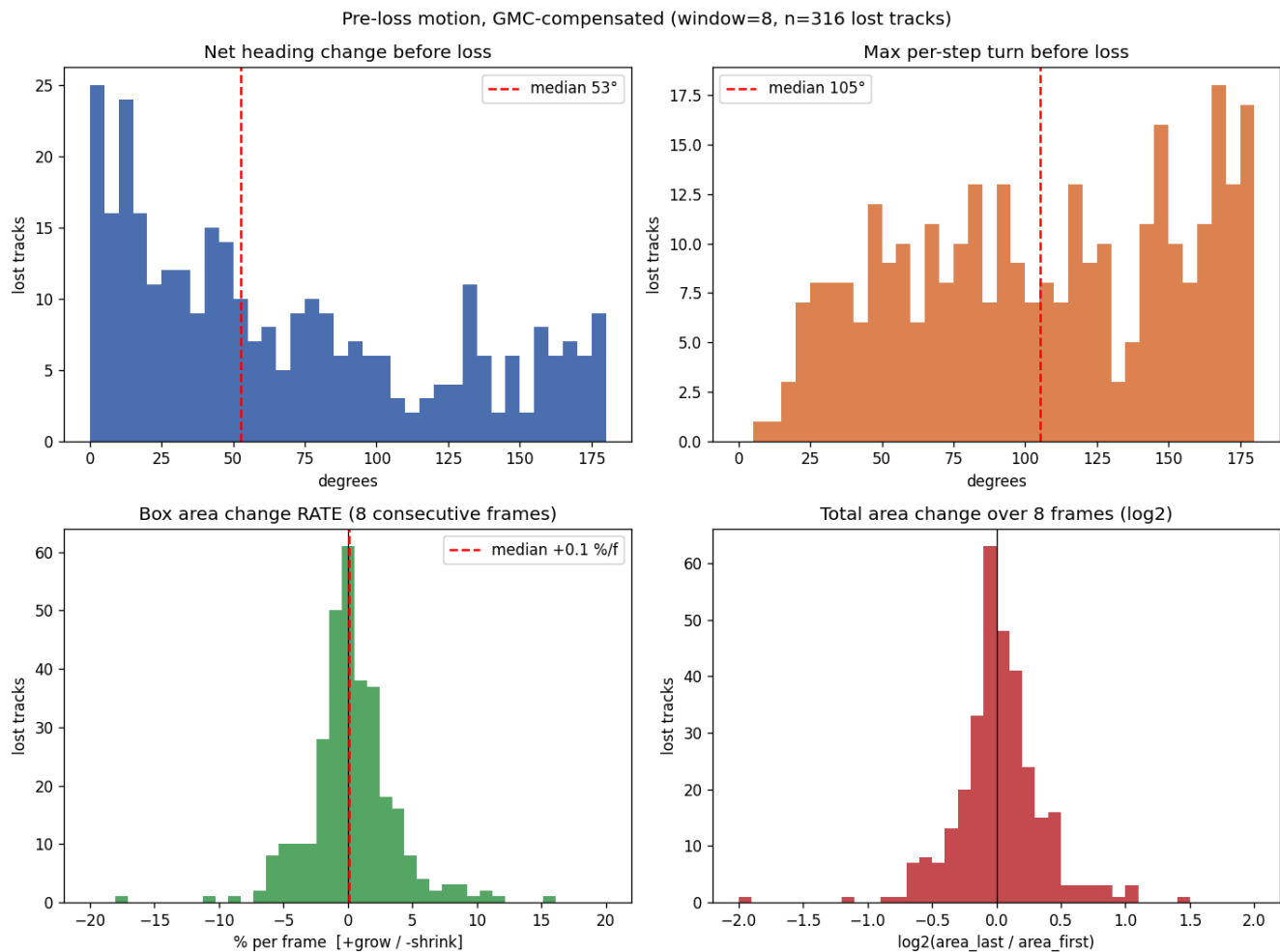
thr (<code>bridge_dist</code> ≤)	TP	FP	prec	recall
0.15	23	40	36.5%	9.0%
0.30	56	161	25.8%	21.9%
0.50	93	425	18.0%	36.3%
1.00	155	1355	10.3%	60.5%

- **The 0.895 is mostly easy far-away negatives.** Recomputing `bridge_dist` AUC on progressively tighter (spatially plausible) pools: `dist_h≤1` →0.826, `dist_h≤0.6` →0.813, but in the **gate's operating region** `bridge_dist≤1` →**0.675** and `bridge_dist≤0.5` →**0.644**. Among genuine near look-alikes the gate is only marginally better than chance.

Conclusion: geometry is a strong *ranker for rejecting impossible candidates* but is **weak on the hard near cases** (~0.65) and cannot carry precision against a 98.7%-negative pool. The lever is **shrinking the negative pool first** (tighter gap/spatial gate, or appearance/ReID as the first gate) and using geometry only for coarse reachability.

4. Pre-loss kinematics (GMC-compensated)

Per-frame GMC affine (`SparseOpticalFlowGMC` , LK partial-affine) is composed into a world→frame transform; every box is back-projected to a stabilized world frame before measuring motion, so camera pan/rotation/zoom is removed. Tracks lost mid-sequence only.

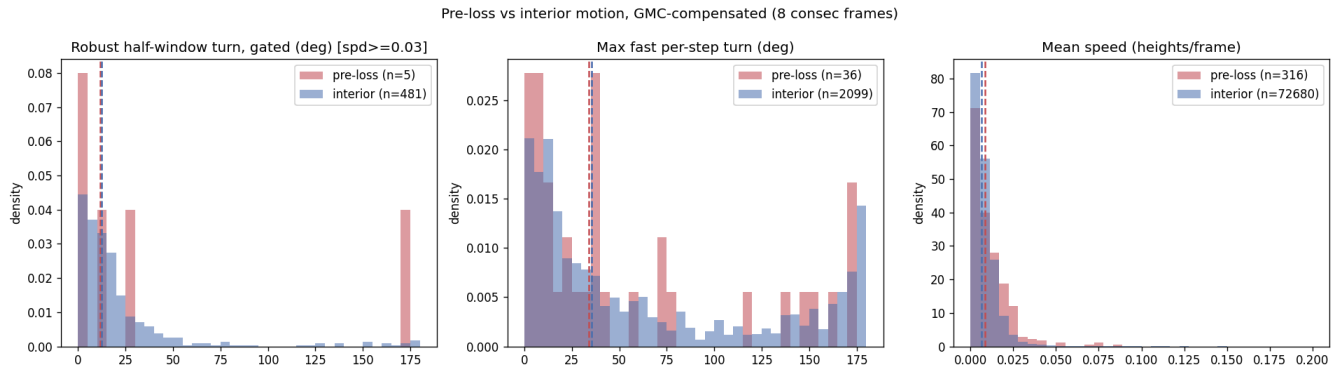


Box area is a robust gate. Over the last 8 *consecutive* frames (316 lost tracks), the area change rate (log-linear fit) has **|median| $\approx$ 1.6 %/frame**, p10-p90 = -3.5...+4.2 %/frame, 8-frame cumulative 0.78–1.48 $\times$, symmetric (no systematic grow/shrink before loss). $\Rightarrow$ a tight relink scale tolerance ($\approx \pm 6$ %/frame) costs almost no true candidates while cutting scale-jumping false links.

Turning before loss is NOT a signal — it is jitter. Net heading change has median 53°, max-per-step 105° — but these are *identical* (or higher) for interior "cruising" windows of the same tracks. Root cause: MOT17 per-frame foot speed is tiny (median **0.01**

heights/frame

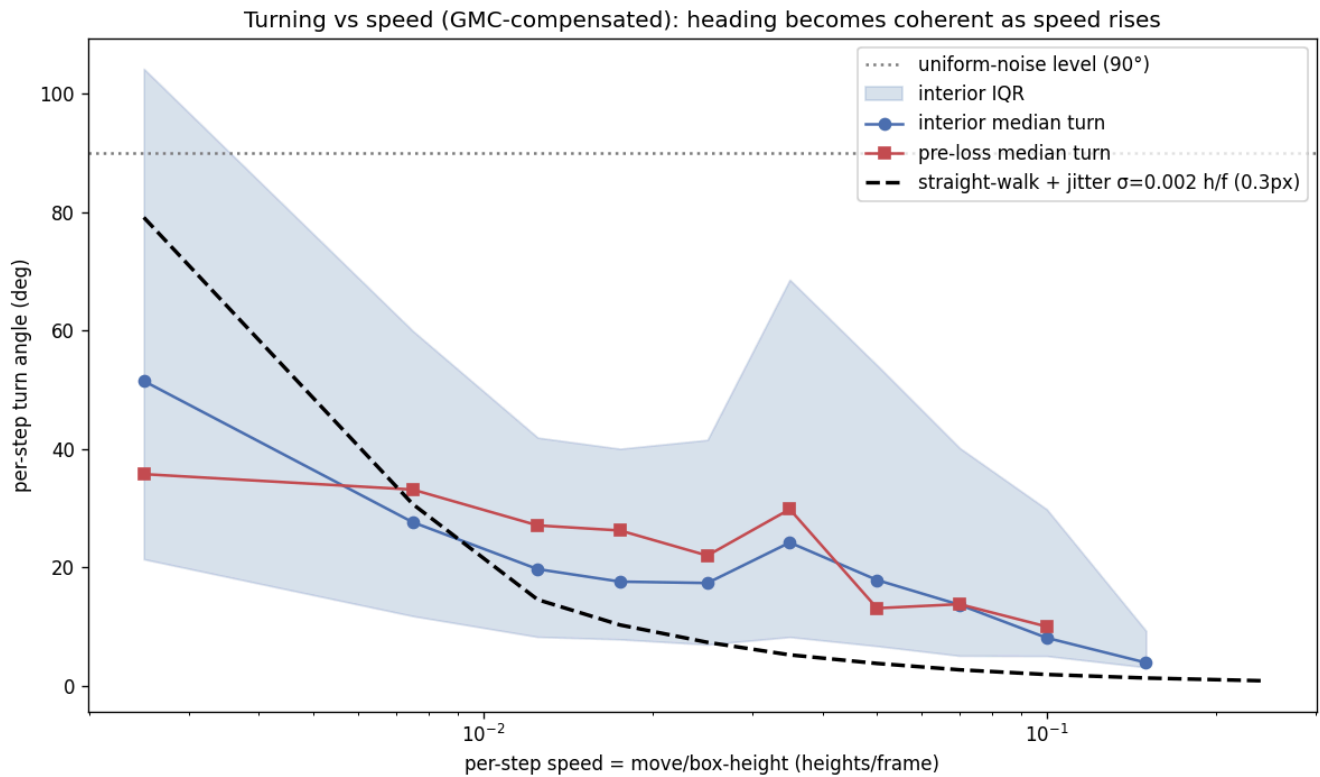
≈ 2 px), below the box-jitter floor, so the velocity *direction* is noise.



Gating per-step turns at speed ≥ 0.03 h/f leaves only **1-2 % of windows**; among those the robust half-window turn collapses to **$\sim 12^\circ$** (interior $\approx$ pre-loss). People walk **near-straight**; constant-velocity extrapolation is fine *directionally* provided velocity is estimated over a multi-frame window, not per-frame. (An earlier in-analysis claim that "sharp pre-loss turning breaks the bridge" is **retracted** — the opposite holds.)

5. Speed $\times$ turning sweep + the underlying distribution

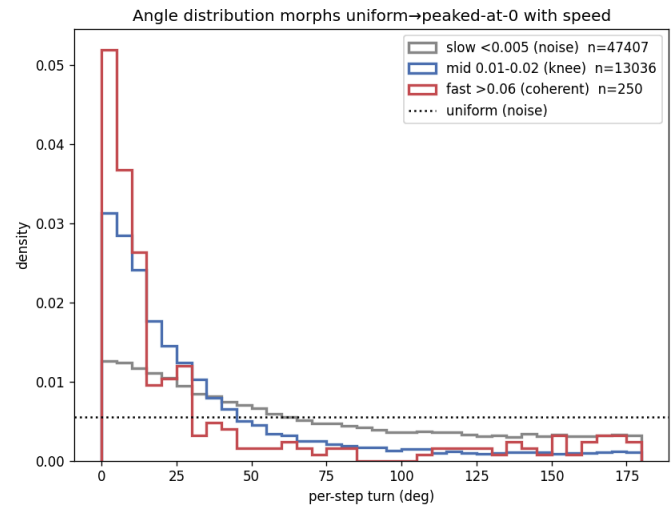
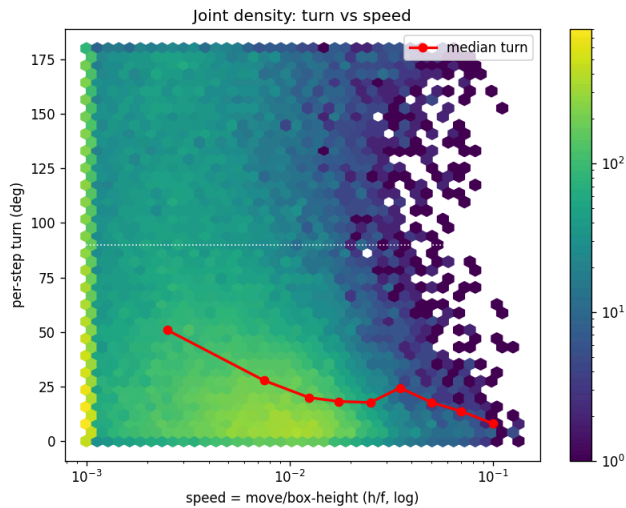
Pooling all 84,163 GMC-compensated 3-frame triplets and binning by per-step speed
(= move / box-height):



speed (h/f)	frames per body-height	n (interior)	median turn
<0.005	>400	45,902	51.5° ($\approx$ noise)
0.01-0.015	80	8,704	19.7°
0.02-0.03	40	2,245	17.3°
0.08-0.12	10	52	8.1°
0.12-0.18	6.7	10	3.9° ($\approx$ straight)

Heading becomes usable at ≈ 0.01 - 0.02 h/f (~ 2 - 4 px/frame for a 169 px-median person; one body-height per 40-80 frames). Below ~ 0.005 h/f (57 % of all steps) it is noise. Only ~ 3.4 % of steps clear 0.02 h/f, so heading helps only the fast-moving minority; the slow bulk must rely on spatial proximity + area + appearance. **Pre-loss $\approx$ interior at every matched speed**

$\Rightarrow$ turning carries no extra pre-loss discriminability.



Distribution family: speed-conditioned projected-normal (offset-normal), $\kappa \propto \text{SNR} = \text{speed}/\sigma$.

The turning angle is not uniform even when slow — it is forward-biased at all speeds and sharpens with speed:

regime	P(turn < 30°)	P(turn > 150°)	median
slow <0.005	0.34	0.10	50.9°
mid 0.01–0.02	0.64	0.03	19.5°
fast >0.06	0.74	0.08	11.8°

(uniform would give $P(<30^\circ)=0.167$.) Speed itself is heavily right-skewed (log-normal-like,

median 0.004 h/f). The pooled marginal turn is therefore a **scale-mixture** of projected-

normals — a 0° peak with a heavy tail to 180° — which is why the un-binned median (~53°) *looks*

uniform but is not. A single- σ "straight walk + Gaussian foot-jitter" Monte-Carlo only roughly

matches (weighted RMSE $\approx 21^\circ$, fit pinned at the σ floor), confirming the mechanism is

near-straight motion blurred by $\sim$ pixel-scale position noise, plus mild real persistence.

6. Implications for relink design

1. **Negative-pool reduction is the lever**, not a better geometric residual.
Tighten gap +
spatial gate to raise base rate, and/or use appearance/ReID as the first gate;
let
`bridge_dist` confirm.
2. **Adopt the area-rate gate** ($\approx \pm 6$ %/frame, or $\leq \sim 1.5\times$ over the gap): near-free precision,
negligible true-candidate loss.
3. **Make heading speed-adaptive**. Use the projected-normal $\kappa(\text{speed})$ as the directional
tolerance: widen the `dir_cos` gate for slow targets (direction unreliable),
tighten it for
fast ones. Estimate velocity over a multi-frame anchor, never per-frame.
4. Heading is irrelevant for the slow majority — do not gate them on direction at all.

6b. Reach-gate model: velocity drift is dead weight (tested & rejected)

Tested the intuition `reach R(G) = vector-length + search-distance` with
`vector-length = s · G` (lost exit speed × gap). `validate_reach_gate.py` ranks each
formulation against `bridge_dist` / plain spatial `dist_h` on the GT-valid pairs:

gate	AUC	recall @ FP 100/500/2000
<code>bridge_dist</code> (online)	0.895	16 / 39 / 68 %
<code>dist_h</code> (constant radius, no motion)	0.868	13 / 39 / 64 %
additive <code>dist_h - s · G</code>	0.504	0 / 0 / 1 %
<code>(dist_h - s · G) / √G</code>	0.515	0 / 0 / —
inverse-weighted <code>(dist_h - s · G) · s · G</code>	0.490	0 / 0 / 0 %

Every formulation that uses the velocity vector collapses to chance (or below — fast subset goes anti-correlated, AUC 0.40). Root cause is the section-4/5 finding: per-track exit speed `s` is at the jitter floor, so `s·G` is noise × gap, and subtracting/weighting by it destroys the otherwise-good spatial signal. Normalising `dist_h` by any positive power of gap also hurts (`/√G` → 0.795, `/G` → 0.673): true relinks stay spatially close regardless of gap, so a **constant search radius beats a growing one**.

Verdict: an *additive/multiplicative* drift term `s·G` carries zero information — `s` is at the jitter floor so `s·G` is noise. Reappearance position is dominated by **where the track vanished (spatial persistence)**, not blind velocity extrapolation.

6c. Speed-weighted velocity term — the one geometric gain that survives

§6b kills the *additive* drift, but the **velocity term earns its keep when, and only when, the target is actually moving**. Per-speed AUC of `bridge_dist` vs pure-spatial `dist_h` (Δ = velocity contribution), split by `min(lost_exit, cand_entry)` speed:

min speed (h/f)	AUC bridge	AUC dist_h	Δ (velocity term)
<0.01 (slow)	0.908	0.911	−0.002
0.01–0.02	0.882	0.865	+0.017
0.02–0.05	0.883	0.815	+0.068
≥0.05 (fast)	0.927	0.846	+0.081

The velocity term's value rises monotonically with speed. Crucially, in the **gate operating**

region (`bridge_dist ≤ 1`) plain `dist_h` (0.775) **beats** `bridge_dist` (0.675) — the online bridge **over-trusts velocity for slow/mid targets and injects noise**. A speed-weighted blend $w(s) \cdot (\text{velocity score}) + (1-w) \cdot \text{dist}_h$, with $w = \text{clip}(\text{min_speed} / 0.05, 0, 1)$, recovers it:

score	AUC full	AUC hard (<code>bd≤1</code>)
<code>bridge_dist</code> (online)	0.895	0.675
<code>dist_h</code> (spatial only)	0.868	0.775
<code>blend(bridge , dist_h)</code>	0.901	0.790
<code>blend(fwd_resid , dist_h)</code>	0.887	0.796

Design rule: always-on spatial proximity (`dist_h`) as the base, **plus** a velocity-consistency term weighted by `w(s)` that grows with speed. Fast targets use the **full** extrapolation, not the capped `gap/2` midpoint.

Optimised config (`optimize_relink_weight.py` , grid over residual × speed metric × saturation × shape × normalisation, ranked by AP, validated leave-one-sequence-out):

```
score = -( w·sym_fb + (1-w)·dist_h )
  sym_fb      = 0.5·(fwd_resid + bwd_resid)      # symmetric full extrapolation > gap/2 m
  fwd_resid   = ||(lx + vxl·G) - cx0|| / h_ref    # lost fwd-extrapolated to cand head pos
  bwd_resid   = ||cx0 - vxc·G) - lx|| / h_ref    # cand bwd-extrapolated to lost tail pos
  dist_h      = ||(lx,ly) - (cx0,cy0)|| / h_ref   # direct spatial distance, normalised
  h_ref       = avg(ema_h_lost, ema_h_cand)       # EMA box height of both endpoints (s)
  w           = clip( sqrt(s_lost / 0.12), 0, 1 ) # sqrt ramp, saturates ~0.12–0.2 h/f
  s_lost      = ||(vxl, vyl)|| / h_ref           # lost-track exit speed in heights/frame
```

`h_ref` is the **reference scale** that converts all pixel distances into body-height units,

making the threshold `bridge_px` invariant to camera zoom and subject distance. Using the average of the EMA heights of *both* endpoints rather than one alone keeps the normalisation symmetric: if the candidate re-enters the frame closer (larger box), the shared scale still reflects the true scene geometry at the gap midpoint rather than one extreme.

gate	LOSO-CV AP (hard region, held-out)
speed-weighted (per-fold tuned)	0.372 ± 0.144
<code>bridge_dist</code> (online)	0.345 ± 0.142
<code>dist_h</code> (spatial only)	0.201 ± 0.143

The config is stable (5/7 folds pick `sym_fb`, `s_lost`, `sat≈0.2`, `sqrt`). **Honest offline gain is +0.027 AP (~+8 %)** — the in-sample AP (0.16→0.32) is optimistic; cross-validation is the real number. Three data-chosen design points: (1) symmetric `0.5(fwd+bwd)` residual beats the one-sided midpoint; (2) `sqrt` weight with a **high** saturation (~0.15 h/f) — use *sparingly*, only for clearly-fast tracks; (3) no normalisation needed.

6d. Online validation — the gain is real end-to-end

Wired into the GPU tracker-core bridge (`relink_bidir_propose_kernel` in `src/tracking/tracker_gpu.cu` + the CPU `midpoint_bridge_dist` in `tracker_gpu_python.cpp`) as the **sole** foot-bridge score — the legacy gap/2 midpoint is removed, not flagged (it is strictly worse and preserved in git history + the A/B below). The bridge stays default-off, so

the tracking baseline is unaffected. `--relink-bridge-px` default is now **0.25**.
MOT17 train
SDP, `mamba_whole_graph`, `--relink-bridge-enabled`, px swept:

config	IDF1	HOTA	AssA	IDs
relink OFF	73.0	66.5	63.8	558
bridge, legacy midpoint (px 0.3)	74.2	67.5	65.2	497
bridge, speed-weighted (px 0.25)	74.8	68.0	66.2	483

px curve peaks on a 0.25–0.30 plateau (74.7–74.8); ≤ 0.2 over-tightens (rejects true bridges, IDs rise), ≥ 0.4 over-admits. **Speed-weighted vs legacy bridge: IDF1 +0.6, HOTA +0.5, AssA +1.0, IDs –14** — above the ± 0.3 pp noise band, and AssA (the HOTA bottleneck) is where it lands.
mode-0 re-run reproduces 74.2 exactly (bit-identity confirmed). The offline analysis translated cleanly to an online win; remaining headroom is base-rate (appearance).

6e. h_ref regression and revert (2026-06-11)

After §6d was validated, commit `b605ac3e` changed `h_ref` from `avg(ema_lost, ema_cand)` to `ema_lost` only across all three bridge paths (GPU kernel, C++ semantic relinker, Python relinker).
The stated rationale was to prevent the candidate's size from inflating the gate when the two tracks differ in scale. However this regressed online performance:

h_ref formula	IDF1	HOTA	AssA	IDs
avg (original)	74.8	68.0	66.2	483

h_ref formula	IDF1	HOTA	AssA	IDs
lost-only (b605ac3e)	74.4	67.6	65.5	495

Root cause: `bridge_px = 0.25` is a normalised threshold; switching from avg to lost-only shifts

the effective gate asymmetrically. When the candidate is *larger* than the lost track (person

walking toward camera during the gap), `h_ref` shrinks → normalised distances grow → some true

bridges are incorrectly rejected. When the candidate is *smaller*, `h_ref` grows → distances

shrink → the gate becomes slightly looser. The net effect is a precision drop (IDs +12) without

any recall benefit.

The "inflate" concern is also made redundant by the scale gate introduced later (§6f): if the

lost/cand height ratio falls outside `[0.75, 1.33]` the pair is already rejected before the

distance is computed.

Revert: `b605ac3e` was reverted in all three paths (`tracker_gpu.cu` , `relink_gate.cu` , `relink.py`). Golden gate (`eval_golden.py check`) passes bit-exactly on the default-off path.

6f. Precision gate ablation (2026-06-11)

Three gates were tested on the speed-weighted bridge (`h_ref = avg`, `bridge_px = 0.25`):

Scale gate (`bridge_h_lo / bridge_h_hi`): rejects pairs whose lost/cand EMA-height ratio

falls outside `[h_lo, h_hi]` . Motivated by §4: 8-frame cumulative scale is 0.78-1.48×, so a

symmetric $\pm 33\%$ tolerance [0.75, 1.33] matches the natural pre-loss range and cuts large

size-jumping false links. Offline simulation on live-accepted pairs: kills **53 % of false**

bridges at zero short-gap TP loss (rejected TPs all have gap ≥ 37 frames).

Implemented in all
three bridge paths.

Margin (`bridge_margin`): rejects the best candidate when

`second_best_dist - best_dist < margin` ,

i.e. the match is ambiguous. Prevents committing to a bridge in crowded scenes where two lost

tracks are nearly equidistant. Analogous to `reciprocal_margin` in the semantic relinker.

Spatial pre-filter (`bridge_spatial_gate`): rejects pairs whose centre distance / `h_ref`

exceeds the threshold *before* computing the velocity-extrapolated score. Intended as a cheap

coarse filter to skip obviously-impossible pairs.

Online ablation results (MOT17 train SDP, `mamba_whole_graph` , `h_ref` = avg, all gates isolated

then combined):

config	IDF1	HOTA	AssA	IDs	FP	FN
bridge, speed-weighted (<code>h_ref</code> avg)	74.8	68.0	66.2	483	3541	20994
+ scale gate [0.75, 1.33] only	74.9	68.0	66.2	482	3516	21056
+ margin = 0.05 only	75.1	68.2	66.6	483	3528	21040
+ spatial_gate = 3.0	74.8	68.0	66.2	483	3541	20994
+ scale + margin = 0.05 (combined)	75.1	68.2	66.6	482	3514	21082

Key findings:

- **Margin = 0.05** is the dominant gate with avg h_{ref} (+0.3 IDF1, +0.4 AssA, FP −13). Rejecting ambiguous bridges (second-best within 0.05 of best) directly reduces false relinks in crowded scenes. Values beyond 0.05 give identical metrics — the ambiguous-pair population is exhausted.
- **Scale gate [0.75, 1.33]** contributes a smaller but independent gain on top (IDs −1, FP −14 vs margin-only). With avg h_{ref} the gate's marginal value is lower than with lost-only h_{ref} : avg normalization already partially penalises size-mismatched pairs, so fewer pairs survive to be killed by the explicit ratio check.
- **Spatial pre-filter** (`bridge_spatial_gate = 3.0` or `5.0`) is **bit-identical** to the baseline. `bridge_px = 0.25` is already a tight gate; any pair that passes the speed-weighted score also has a centre distance well within a few `h_ref` units.
- Combined (scale + margin): negligible over margin-only in IDF1/HOTA/AssA; FP −14 more vs margin-only. Kept as default because the scale gate is near-free and blocks a distinct failure mode (teleporting tracks) that the margin test does not cover.

Defaults set to: `bridge_margin = 0.05` , `bridge_h_lo = 0.75` , `bridge_h_hi = 1.33` . `bridge_spatial_gate` left at 0 (disabled). These are in `lifecycle.py` and `eval/config.py` .

7. Artifacts & reproduction

Output	Produced by
<code>results/MOT17_eval/*.txt</code> (substrate)	<code>mot17.py --preset mamba_whole_graph --detecto</code>
<code>scripts/tools/out/relink_candidates.csv</code>	<code>build_relink_candidates.py</code>

Output	Produced by
figures/preloss_motion.png	analyze_preloss_motion.py --window 8
figures/turn_baseline.png	analyze_turn_baseline.py --min-speed 0.03
figures/speed_turn_sweep.png , out/speed_turn_sweep.npz	sweep_speed_turn.py
figures/speed_turn_dist.png	npz post-plot (see script header)
scripts/tools/out/reach_gate.png	validate_reach_gate.py
speed-weight grid + LOSO CV (stdout)	optimize_relink_weight.py